

Operating System Midterm (2025)

Question Paper ID: A

Name :

Student ID :

1. Question Paper ID: ____ .
2. Please explain the following terms: (1) **Interrupt** (3%) (2) **Context Switch** (4%) (3) **Program Control Block**. (3%)
3. Please explain what a **Timesharing** system is (5%) and explain why time slices are usually set to be very short. (5%)
4. Please give an example from operating systems to explain the difference between **mechanism** and **policy** in operating system design. (10%)
5. What data is stored in the **text section**, **data section**, **heap section**, and **stack section** respectively? (10%)
6. Please explain the basic concepts of the two main communication models in **IPC** (Inter-Process Communication). (10%)
7. Please explain the difference between **data parallelism** and **task parallelism**. (10%)
8. Sleep Sort is a sorting algorithm that works by spawning one thread (or process) per number. Each thread sleeps for an amount of time proportional to its number, then prints the number when it wakes up. Because smaller numbers wake up first, the output appears in sorted order. Explain why using very short sleep durations may cause Sleep Sort to fail. You may use an example to support your explanation. (10%)
9. (1) In CPU Scheduling, please explain what is the **Convoy Effect**? (3%) (2) Please give an example of a scheduling algorithm that can cause the Convoy Effect. (3%) (3) In the answer to the previous question, is that algorithm Nonpreemptive or Preemptive? (4%)

10. Given the following set of processes with their arrival times and CPU burst times:

Process	Arrival Time	Burst Time
P1	0	6
P2	2	3
P3	3	7
P4	5	2

- (a) Calculate the waiting time for each process using First-Come, First-Served (FCFS) scheduling. (2%)
- (b) Calculate the waiting time for each process using Non-preemptive Shortest-Job First (SJF) scheduling. (2%)
- (c) Calculate the waiting time for each process using Preemptive Shortest-Job First (SJF) scheduling. (3%)
- (d) Calculate the waiting time for each process using Round-Robin (RR) scheduling with a time quantum of 4 units. (Hint: When a process's time quantum expires or a new process arrives, that process is inserted at the end of the ready queue.) (3%)

11. Consider the following set of *periodic real-time tasks*. All tasks are released simultaneously at time 0.

Process	Period = Deadline	Execution Time
P1	30	13
P2	50	25

- (a) Draw the Gantt chart for Rate-Monotonic Scheduling (RMS)(4%). Does RMS meet all deadlines? (1%)
- (b) Draw the Gantt chart for Earliest-Deadline-First Scheduling (EDF) from time 0 to time 100 (4%). Does EDF meet all deadlines? (1%)